

e-values for multiplicity control in multiverse analysis

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Abstract

Multiverse analysis refers to a common situation in the Social Sciences where one wishes to assess the association between multiple potential treatment definitions and multiple potential outcome definitions, possibly within multiple sub-populations. To avoid a selective reporting of results, it is critical to acknowledge that multiple comparisons are being performed, and to ensure a valid statistical control of false positive findings. We propose the use of e-values as a tool to guarantee that the false discovery rate stays within a user-specified level, while accounting for the effect of confounding covariates. We illustrate the application of our methodology on a dataset studying the association between technology use (defined in multiple ways) on well-being (also defined in multiple ways).

Keywords: multiverse analysis, e-value, multiple testing, false discovery rate

1 Introduction

There are many applications where one wants to assess the effect of multiple treatments on multiple outcomes, potentially also separately within multiple sub-populations. A prominent example in the Social Sciences occurs when one has multiple outcomes that

can be conceivably thought of as measuring related underlying characteristics, such as different measures of psychological well-being, several measures of economic activity, or several measures of radicalization in social media usage. Similarly, the multiple treatments may refer to using various types of technology, realizing various economic interventions, or exposure to various social media unradicalization campaigns. It may also be natural to wish to assess treatment effects within sub-populations, for example defined by sex and age groups, economic regions, or social media user backgrounds. Beyond the Social Sciences, similar situations arise in other fields. In Biomedicine one may study the effect of non-identical, but related, treatments on multiple measures of disease progression, in Climate Science one may study the association between multiple human-driven processes on multiple physical parameters, etc.

Multiverse analysis aims to provide a principled data analysis framework to consider such multiple outcome-treatment-subpopulation combinations. A key pitfall to avoid is selective reporting, whereby researchers obtain treatment effect estimates for all combinations and report only those that are larger. Critically, selective reporting occurs when a researcher fails to indicate all the multiple comparisons that were made. A strategy to avoid this pitfall is to report all the analyses that were conducted, and account for the multiple comparisons when assessing the statistical significance of the findings. This is the strategy followed in this paper.

Before proceeding, we discuss another popular strategy: focus the analysis on some summary measure across all the comparisons being made. This is the standard approach for meta-analyses in the medical literature, where one combines the evidence regarding the effect of a medical treatment (or family or related treatments) across studies (typically, randomized clinical trials). Therein, it is common to account for treatment effect heterogeneity, often driven by differences in the patient population targeted by each study. A notable example in the Social Sciences, where randomized trials are rare, is Specification Curve Analysis (SCA, [1]). By default, SCA reports the median estimated effect across treatment-outcome-subpopulations, and assesses the statistical significance of this single parameter. An advantage of using a single summary measure over multiple comparison adjustments is that, since no significance corrections are needed, if all treatment effects are comparable then one may have higher statistical power to detect that the summarized effect truly exists. However, we urge strong caution in the sense that aggregate measures should only be used in appropriate circumstances. A critical pitfall is that, when the treatments are not comparable or there is significant heterogeneity in their effects, the aggregate measure may be highly misleading. For example, [2] showed that SCA led [3] to erroneously conclude that the association between technology use and teenager well-being was mild, practically negligible. The reason was that the association was near-zero for some technologies such as television use, but large for others such as mobile phone use, and hence overall mild on the average. Another critical issue arises when the aggregate measure is obtained across multiple possible subsets of control covariates, an option that is explicitly considered in SCA. This is inadequate because one combines quantities that have a different statistical meaning, such as marginal and conditional associations, and hence the summary has no natural interpretation. For example, assessing the conditional association between salary and sex given an individual job's position, education and

other characteristics has a different meaning than assessing the marginal association between salary and sex. A marginal association between sex and salary can be partially driven by people from different sexes making different career choices, whereas an association between sex and salary within people with the same job position, education and characteristics suggests salary discrimination. ♠ Paul: I think "gender" would be more appropriate than sex See [2] for an in-depth discussion of issues with SCA.

In this paper we propose the application of a simple and practical approach to multiplicity control in multiverse analysis based on e-values. Briefly, e-values are an increasingly popular approach to hypothesis testing that has several features relative to P-values. ♠ Paul: should we write "p-value" since we write "e-value"? When comparing two hypotheses, e-values accumulate evidence against the null hypothesis similarly to a Bayes factor (see [4] for a review). Further, e-values guarantee a strict control of the type I error probability, and are particularly suited to control the false discovery rate (FDR) in multiple comparisons [5]. Although less relevant to this work, e-values also enable anytime-valid inference, a strong asset in settings where one accumulates data and makes decisions sequentially. For seminal papers on e-values see [6–8], and for an excellent monograph see [9].

We briefly review the rich literature on FDR control for multiple testing. [10] proposed that, when conducting many statistical tests, one should use procedures that guarantee controlling the FDR below a user pre-specified level α . The FDR is the expected proportion of false positives among the hypotheses that are claimed as statistically significant. [10] proposed a P-value adjustment procedure that controls the FDR under the assumption that the tests are independent, and [11] a variation that assumes that the tests follow a type of positive dependence structure. A large body of subsequent work extended the FDR framework in several directions. A prominent example are the q-values of [12]. Analogously to p-values controlling the expected number of false discoveries, q-values control the expected positive FDR, that is the expected fraction of false discoveries, conditional on ≥ 1 discoveries being reported. However, q-values require estimating nuisance parameters such as the proportion of true null hypotheses, and therefore do not control the FDR exactly. Further, estimating said nuisance parameters often requires simplifying assumption, such as independence between the test statistics. Another influential line of work is based on the local FDR [13], which is embedded into an empirical Bayes framework. The local FDR can be interpreted as the posterior probability that a given hypothesis is null, given the value of the test statistic for that hypothesis. The local FDR provides a more refined, hypothesis-specific, assessment of significance than tail-area measures such as q-values. Estimating the local FDR accurately requires the number of hypotheses being tested to be large (at least in the hundreds), assuming independence across tests, and making certain choices when estimating the global mixture distribution of the test statistics. Further related work includes the optimal discovery procedure of [14], and the Bayesian discovery procedure of [15]. See [16] for a monograph on FDR control.

An advantage of e-values over these methods are that they provide a finite- n FDR control that does not require estimating nuisance parameters, and which holds regardless of the dependence structure between the multiple tests. This is relevant in multiverse analysis, where typically tests cannot be assumed independent. Another

advantage of e-values for multiverse analysis is that they are more composable, for example they can be multiplied or averaged across independent datasets (corresponding to subpopulations where a treatment effect is tested, for example) to obtain valid e-values. A main limitation of e-values relative to methods based on p-values, q-values and empirical Bayes is that the latter usually have a higher statistical power, provided that their assumptions hold. Therefore, our framework is most appropriate for datasets where one has relatively large sample sizes. Otherwise, while FDR control is ensured, one may obtain few statistically significant findings. We conduct a brief simulation study comparing the power of different e-variables applicable in our framework.

The paper is structured as follows. In Section 2 we introduce notation and lay out a generalized linear model framework for treatment effect estimation within the context of a multiverse analysis. Section 3 reviews e-values, issues related to statistical power and discusses their use for multiplicity control. Section 4 applies our methodology to a dataset studying the association between teenager mental well-being and technology use. Section 5 concludes.

2 Framework

We lay out a framework that closely resembles that of [2] but provides finite sample FDR control guarantees through e-values. Consider a setting where one has L outcomes, p_x treatments and p_z control covariates, with n observations for each. Said observations may be potentially split across $p_g \geq 1$ subpopulations. Let $y^{(l)} \in \mathbb{R}^n$ for $l = 1, \dots, L$ be the observed outcomes, X the $n \times p_x$ matrix with the treatment allocations, Z the $n \times p_z$ matrix with the control covariates, and G a $n \times p_g$ matrix encoding the subgroup information (if one is not interested in subgroups, then G is omitted). Let x_i , z_i and g_i be the i^{th} row in X , Z and G respectively. Throughout, we denote expectations by $\mathbb{E}$, probabilities by $\mathbb{P}$, and by $\mathbb{E}_P$ and $\mathbb{P}_P$ expectations and probabilities when data are generated by a specific distribution P . Further, we denote by p the density (or probability mass) function associated to a distribution P , and by q those associated to a distribution Q .

A generalized linear model for outcome l assumes that

$$F(\mathbb{E}(y_i^{(l)} | x_i, z_i, g_i)) = \beta_0^{(l)} + z_i^T \eta_z^{(l)} + g_i^T \eta_g^{(l)} + \sum_{j=1}^{p_x} (\beta_j^{(l)} + g_i^T \delta_j^{(l)}) x_{ij}, \quad (1)$$

where F is the link function (e.g., the identity for linear regression, or the logit for logistic regression), $\beta_0^{(l)} \in \mathbb{R}$ is the intercept, $\beta_j^{(l)} \in \mathbb{R}$ the average effect of treatment j on outcome l , $\delta_j^{(l)}$ captures subpopulation-specific effects, and $\eta_z^{(l)} \in \mathbb{R}^{p_z}$ and $\eta_g^{(l)} \in \mathbb{R}^{p_g}$ are coefficients associated to control covariates and subgroups respectively. We denote the whole parameter vector by $\theta^{(l)} = (\{\beta_j^{(l)}\}_{j=0}^{p_x}, \{\delta_j^{(l)}\}_{j=1}^{p_x}, \eta_z^{(l)}, \eta_g^{(l)})$. Importantly, for this parameter interpretation to hold, g_i must encode subpopulation information using a specific sum-to-zero constraint described next. Let ρ_k the proportion of observations

coming from subpopulation k , we define

$$g_{ik} = \begin{cases} \rho_k, & \text{if observation } i \text{ comes from subpopulation } k \\ -(1 - \rho_k), & \text{otherwise} \end{cases}$$

for $k = 1, \dots, p_g$, which implies that $\sum_{i=1}^n g_{ik} = 0$. We define the effect of treatment j on outcome l for individual i in group $g_i = k$ as

$$\beta_{jk}^{(l)} = \frac{\partial}{\partial x_{ij}} F(\mathbb{E}(y_i^{(l)} \mid x_i, z_i, g_i = k)) = \beta_j^{(l)} + g_i^T \delta_j^{(l)} = \beta_j^{(l)} + \rho_k \delta_{jk}^{(l)} - \sum_{m \neq k} \delta_{jm}^{(l)} (1 - \rho_m).$$

The corresponding average treatment effect (ATE) across all individuals is

$$\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial x_{ij}} F(\mathbb{E}(y_i^{(l)} \mid x_i, z_i, g_i = k)) = \beta_j^{(l)} + \frac{1}{n} \left(\sum_{i=1}^n g_i^T \right) \delta_j^{(l)} = \beta_j^{(l)}.$$

We consider a multiverse analysis where one is interested in estimating the ATEs $\beta_j^{(l)}$ for $l = 1, \dots, L$ and $j = 1, \dots, p_x$, and performing the corresponding Lp_x hypothesis tests

$$\begin{aligned} H_{lj} : \beta_j^{(l)} &= 0, \beta_{j'}^{(l)} \in \mathbb{R} \text{ for } j' \neq j \\ \overline{H}_l : (\beta_1^{(l)}, \dots, \beta_{p_x}^{(l)}) &\in \mathbb{R}^{p_x} \end{aligned} \quad (2)$$

That is, the null hypothesis H_{lj} is that treatment j has zero effect on outcome j , without any restrictions on the remaining treatment effects. And there is a global alternative hypothesis $\overline{H}_l$, shared across all treatments, that treatment effects are non-zero. We denote by $\mathcal{H}_{lj}$ the set of probability distributions such that H_{lj} holds, and by $\overline{\mathcal{H}}_l$ the set where none of $H_{l1}, \dots, H_{lp_x}$ hold. In the event that one is also interested in testing the effect within subpopulations, there are $Lp_x p_g$ further tests corresponding to $H_{ljk} : \beta_{jk}^{(l)} = 0$ for $l = 1, \dots, L$, $j = 1, \dots, p$ and $k = 1, \dots, p_g$. To ease notation we drop the subscript k in the exposition below, that is we describe the case where there are no subpopulations, but our framework remains readily applicable when sub-populations are also of interest.

Although not stated explicitly, no constraints are specified on any of the other elements of $\theta^{(l)}$, neither under the null nor the alternative. In particular, the control covariates effects η_z are always assumed to be non-zero. This is contrast to SCA [1] or its Bayesian counterpart [2], where one also considers covariate subsets. As discussed in Section 1, SCA computes a median over subsets of control covariates that has no statistical interpretation. Its Bayesian counterpart computes a weighted average where weights are given by posterior model probabilities, which as $n \rightarrow \infty$ converges to performing inference under a model that includes only the controls that have non-zero effects. While this implies an asymptotic gain in statistical power, one does not obtain strict type I error control for finite n . Hence our strategy here of including all controls, as an alternative that is more conservative but strictly controls false positives.

Given the potentially large number of multiple tests being conducted, it is important to assess statistical significance in a way such that some measure of false positives is controlled.

Specifically, we consider the FDR [10]. Let $\gamma_{jkl} = \mathbf{I}(\beta_{jk}^{(l)})$ be the indicator that the null hypothesis doesn't hold H_{lj} , let $\hat{\gamma}_{lj} = 1$ if we reject H_{lj} and $\hat{\gamma}_{lj} = 0$ otherwise. Then $\text{FDR} = \mathbb{E}(\text{FDP})$, where

$$\text{FDP} = \begin{cases} \frac{\sum_{jkl} \gamma_{lj} \hat{\gamma}_{lj}}{\sum_{jkl} \hat{\gamma}_{lj}}, & \text{if } \sum_{jkl} \hat{\gamma}_{lj} > 0 \\ 0, & \text{otherwise} \end{cases}$$

and the expectation is with respect to the data-generating distribution. That is, the FDR is the expected false discovery proportion (FDP) and, if no hypotheses are rejected, then FDP is defined to be 0.

3 e-values

E-values provide a convenient way to assess statistical significance in a way that the FDR is strictly controlled, for any sample size and any dependence structure between the tests being performed. Formally, an e-value is the realization of a random variable called an e-variable, whose defining property is that its expectation under the null hypothesis is ≤ 1 .

Definition 1 Consider a null hypothesis H_{lj} specifying a set of probability distributions $\mathcal{H}_{lj}$. A test statistic E_{lj} is an e-variable if and only if $\mathbb{E}_P(E_{lj}) \leq 1$ for any $P \in \mathcal{H}_{lj}$.

Larger values of E_{lj} provide stronger evidence for the alternative hypothesis. By Markov's inequality, if one rejects H_{lj} when $E_{lj} \geq 1/\alpha$, then the type I error is $\leq \alpha$:

$$\mathbb{P}_P \left(E_{lj} \geq \frac{1}{\alpha} \right) \leq \alpha \mathbb{E}_P(E_{lj}) \leq \alpha,$$

for any $P \in \mathcal{H}_{lj}$.

In Section 3.1 we discuss strategies to define e-variables applicable in our generalized linear model framework. In Section 3.2, we conduct a simulation study assessing the statistical power of a set of e-variables in linear and logistic regressions. Section 3.3 reviews e-value-based procedures for FDR control.

3.1 Defining e-variables

There are many ways to define valid e-variables. While the Markov inequality guarantees type I error control for all e-variables, their statistical power may vary greatly. Efforts have been made to define e-variables that accumulate evidence against the null as fast as possible. Such optimality is usually defined in terms of maximizing $\mathbb{E}_Q[\log E_{lj}]$, where $Q \in \overline{\mathcal{H}}_l$ is a distribution within the global alternative hypothesis.

One possible definition of optimality is the so-called numeraire, an e-variable E_{lj}^* ([9], Chapter 6) that is defined by E_{lj}/E_{lj}^* having expectation ≤ 1 under the alternative, where E_{lj} is any other e-variable. Another possible definition by [7] uses a worst-case optimality criterion called GROW. When testing a point null hypothesis versus a point null alternative, both definitions return the likelihood-ratio test as the optimal e-variable. This is of course not applicable to our setting, where the null hypothesis $\mathcal{H}_{lj}$ contains infinitely many distributions indexed by $\theta^{(l)}$, and so does the alternative $\overline{\mathcal{H}}_l$. In such more general settings, obtaining either the numeraire or the GROW-optimal e-variable is related to so-called reverse information projections (RIP), and it is a highly non-trivial task.

We review next some practical alternatives applicable to our generalized linear model framework (1). A first option is based on universal inference introduced by [17]. Specifically, one splits the observed data $y^{(l)} = (y_0^{(l)}, y_1^{(l)})$, where $y_0^{(l)}$ is assumed independent from $y_1^{(l)}$. One then obtains the maximum likelihood estimate (MLE) for the density function under the null $\hat{p}_{lj} \in \mathcal{H}_{lj}$ using only $y_0^{(l)}$, that under the global alternative $\hat{q}_l$ using only $y_1^{(l)}$, and the split likelihood-ratio e-variable proposed in [18] is defined as

$$E_{lj} = \frac{\hat{q}_l(y_0^{(l)})}{\hat{p}_{lj}(y_0^{(l)})}.$$

A variation of split likelihood-ratio e-variables discussed in [9] (Chapter 5.2), the universal mixture e-variable, avoids using arbitrary data splits (as well as assuming independence) by considering a hybrid of frequentist and Bayesian hypothesis tests. Specifically, let $\hat{p}_{lj} \in \mathcal{H}_{lj}$ be the MLE under the null, now using the full data $y^{(l)}$. Define

$$E_{lj}^{um} = \frac{q_l(y^{(l)})}{\hat{p}_{lj}(y^{(l)})}, \quad (3)$$

where

$$q_l(y) = \int p(y \mid \theta^{(l)}) p(\theta_l \mid \overline{\mathcal{H}}_l) d\theta_l$$

is the marginal (or integrated) likelihood with respect to a prior density $p(\theta_j^{(l)} \mid \overline{\mathcal{H}}_l)$ that places full support on the alternative (in our setting, under $p(\theta_j^{(l)} \mid \overline{\mathcal{H}}_l)$ the prior probability that any $\theta_j^{(l)} = 0$ is zero). It is easy to show that E_{lj} is an e-variable for any choice of $p(\theta_j^{(l)} \mid \overline{\mathcal{H}}_l)$. Since $\hat{p}_{lj}$ is the MLE under $\mathcal{H}_{lj}$, by definition we have that $p(y^{(l)}) \leq \hat{p}_{lj}(y^{(l)})$ for any $P \in \mathcal{H}_{lj}$, therefore

$$\mathbb{E}_P(E_{lj}) = \int \frac{q_l(y^{(l)})}{\hat{p}_{lj}(y^{(l)})} p(y^{(l)}) dy^{(l)} \leq \int \frac{q_l(y^{(l)})}{p(y^{(l)})} p(y^{(l)}) dy^{(l)} = \int q_l(y^{(l)}) dy^{(l)} = 1.$$

In our framework (1), to avoid subjectivity in the choice of $p(\theta_j^{(l)})$, one may consider Zellner's unit information prior [19], a classical default that has tight connections to

the Bayesian information criterion [20]. Specifically, let $W = (X, Z, G)$ be the design matrix associated to the regression equation in (1) (if subpopulations are of interest, then the interaction terms GX are also included in W). Zellner's unit information prior is

$$\theta_j^{(l)} \sim N(0, n\sigma_l^2(W^T W)^{-1}), \quad (4)$$

where $\sigma_l^2 > 0$ is the GLM dispersion parameter associated to $y^{(l)}$. This choice assumes that $W^T W$ is invertible, as is the case in our examples here, otherwise an alternative default is to replace $W^T W$ by $\text{diag}(W^T W)$. Expression (3) requires obtaining an MLE for the Lp_x alternative hypotheses, which can be done efficiently and in parallel using any suitable optimization algorithm. It also requires obtaining L marginal likelihoods $q_l(y)$, one for each outcome. In the linear regression case where $y^{(l)} \sim N(W\theta^{(l)}, \sigma_l^2 I)$, q_l has a closed-form. Specifically, if one sets an inverse Gamma prior $\sigma_l^2 \sim \text{IG}(a/2, l/2)$, then

$$\frac{q_l(y^{(l)})}{\hat{p}_{lj}(y^{(l)})} = \frac{l^{a/2}\Gamma(\frac{n+a}{2})}{(n+1)^{d/2}\Gamma(a/2)} \frac{(2e/\hat{\sigma}_l^2)^{\frac{n}{2}}}{\left(l + \|y^{(l)}\|^2 - \frac{n}{n+1}(\hat{\theta}^{(l)})^T W^T W \hat{\theta}^{(l)}\right)^{\frac{n+a}{2}}}$$

where $\|\cdot\|^2$ is the squared Euclidean norm, $d = \dim(\theta^{(l)})$, $\hat{\theta}^{(l)} = (W^T W)^{-1}W^T y^{(l)}$ is the least-squares estimator and $\hat{\sigma}_l^2 = n^{-1}(\|y^{(l)}\|^2 - \|W\hat{\theta}^{(l)}\|^2)$ the MLE for the error variance. Outside of linear regression numerical integration methods must be used and, as long as the number of outcomes L is not too large, this can also be done quickly (and in parallel). One may approximate q_l using Laplace approximations, which is very fast as it requires a single optimization exercise for each outcome, and is very accurate as n grows ([21], [22] Lemma S3).

A second option is a soft-rank e-variable studied in [23]. A soft-rank e-variable is the ratio of a nonnegative test statistic for the test of interest, T_0 , and the average of B others statistics for the test, $T_1, \dots, T_B$, that are exchangeable together with T_0 under the null,

$$E^{sr} = \frac{(B+1)T_0}{\sum_{b=0}^B T_b} \quad (5)$$

The assumed exchangeability guarantees that E^{sr} is indeed an e-variable. In our generalized linear model framework (1), T_0 , can be taken to be the likelihood ratio test statistic for (2), that is,

$$T_0 = \frac{\hat{q}_l(y^{(l)})}{\hat{p}_{lj}(y^{(l)})} \quad (6)$$

where, recall, $\hat{p}_{lj}$ is the maximum likelihood estimate (MLE) for the density function under the null and $\hat{q}_l$ that under the global alternative. Exchangeable likelihood ratio

test statistics $T_1, \dots, T_B$ can be obtained by any permutation method for regression,

$$T_b = \frac{\tilde{q}_l^b(y(l))}{\tilde{p}_{lj}^b(y)}$$

where, $\tilde{p}_{lj}^b$ is the maximum likelihood estimate (MLE) for the density function under the null and $\tilde{q}_l^b$ that under the global alternative, computed over the permuted dataset b . In our simulations, we obtain permuted datasets by permuting residuals of the regression of the treatment variables on the rest of the covariates.

A third option to obtain an e-variable is to use so-called p-to-e calibrators [9] that are class of function that convert any p-value for a test of interest into a valid e-value. Let π_{lj} be a p-value for the test (2), two popular calibrated e-variables [9] are then

$$E^{c1}(\pi_{lj}) = \int_0^1 \kappa \pi_{lj}^{\kappa-1} d\kappa = \frac{1 - \pi_{lj} + \pi_{lj} \log \pi_{lj}}{\pi_{lj} (\log \pi_{lj})^2}, \quad (7)$$

which uses a mixture over $k \in (0, 1)$ of the calibrator function proposed in [24], and

$$E^{c2}(\pi_{lj}) = \pi_{lj}^{1/2} - 1. \quad (8)$$

The family of p-to-e calibrators is however much richer with no single calibrator shown all the others. See [9] (Chapter 2) for a review. In our setting one may take π_{lj} to be the p-value of the likelihood ratio test for (2) with statistic (6). A caveat is that, outside of the Gaussian family, most implementations of the test rely on the asymptotic distribution of (6). The output p-value is then only asymptotically valid, and so is the resulting calibrated e-value. This is the case of the p-value output by R function `anova` from the `stats` package that we use in our simulations and application.

3.2 Simulation study

To assess the power of the three alternatives discussed in 3.1, we run a simulation study in Gaussian (linear) and logistic regressions. We consider sample sizes n ranging from 100 to 10 000 and a fixed dimension $d = 10$. For each family and each sample size, we simulate 100 datasets with data-generating regression coefficients $\beta_i^* \in [0, 1]$, $i = 1, \dots, 10$.

Figure 1 plots the average log e-value over the 100 repetitions for $\beta_i^* \in \{0, 0.15, 0.3, 0.5\}$ in logistic regression. All considered e-variables exhibit good type 1 error control with negative or nearly null log e-values for $\beta_i^* = 0$. For nonzero effects and small n , the calibrated e-values E^{c1} and E^{c2} clearly dominate the universal mixture and soft-rank e-variables E^{um} and E^{sr} , accumulating evidence faster. For large n , the soft-rank e-variable E^{sr} matches the calibrated e-variable E^{c1} , both dominating the universal mixture e-variable E^{um} and the calibrated e-variable E^{c2} .

Figure ?? plots the average log e-value over 100 repetitions for the same β_i^* values as in Figure 1 but in Gaussian linear regression. Findings are similar to that in logistic regression, and, out of the assessed e-variables, we recommend E^{c1} for test (2).

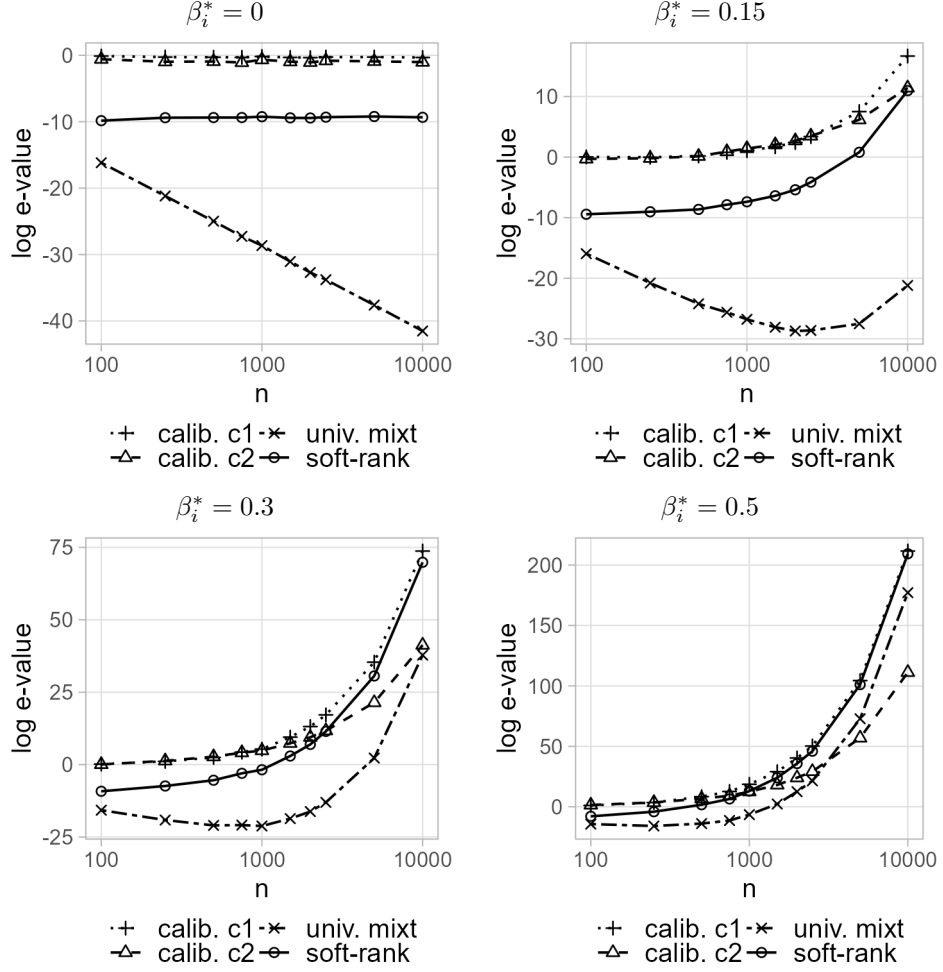


Fig. 1 Average log e-values in logistic regression for $\beta_i^* = 0$ (top left), $\beta_i^* = 0.15$ (top right), $\beta_i^* = 0.3$ (bottom left) and $\beta_i^* = 0.5$ (bottom right).

3.3 FDR control with e-variables

A popular and simple strategy to control the FDR below any user-specified α is the e-BH procedure of [5]. Let e_{lj} be the e-value, that is the observed value of the e-variable E_{lj} , for $l = 1, \dots, L$ and $j = 1, \dots, p_x$, and let $K = Lp_x$ be the number of hypotheses being tested. Let $e_{[k]}$ be the k^{th} order statistic of e_{lj} across all (l, j) , sorted from largest to smallest. The e-BH procedure at level α rejects all hypotheses with the k^* largest e-values, where

$$k^* = \max \left\{ k \in \{1, \dots, Lp_x\} : \frac{ke_{[k]}}{Lp_x} \geq \frac{1}{\alpha} \right\},$$

with the convention that $\max \emptyset = 0$. That is, the k^{th} test effectively uses $ke_{[k]}/K$ as its e-value, and if we reject test k then we also reject all the previous tests. [5] proved that $FDR \leq \alpha K_0/K \leq \alpha$, where K_0 is the number of truly null hypotheses.

A slight refinement of e-BH, which only has a practical effect when K is small, is as follows. Test the joint null hypothesis that all $\theta_{lj} = 0$: if $K^{-1} \sum_{l,j} E_{lj} \leq 1/\alpha$ then no hypothesis is rejected, otherwise apply the e-BH procedure replacing α by $\alpha K/(K-1)$. This procedure controls the FDR below α and dominates uniformly the e-BH procedure. A further improvement is possible by using the closed e-BH procedure ([9], Chapter 9.6), denoted by $\overline{\text{e-BH}}$. Specifically, let R_k the set of indices corresponding to the k largest e-values, $\overline{\text{e-BH}}$ rejects all null hypotheses $k \leq \bar{k}$, where $\bar{k} = \max\{k \geq 0 : R_k \in C_\alpha\}$ and

$$C_\alpha = \left\{ R : \frac{1}{|A|} \sum_{k \in A} E_k \geq \frac{\text{FDP}_A(R)}{\alpha}, \text{ for all } A \subseteq \{1, \dots, K\} \right\}$$

where $\text{FDP}_A(R) = |A \cap R|/\max\{|R|, 1\}$ is the false discovery proportion associated to rejecting the hypotheses in R , if A were the subset of truly null hypotheses. In words, $R_{\bar{k}}$ is such that the average of the e-values over any other subset A exceeds $\text{FDP}_A(R)/\alpha$. $\overline{\text{e-BH}}$ controls the FDR at level α , and rejects at least as many hypothesis as the refined e-BH procedure. A disadvantage however is that finding $R_{\bar{k}}$ requires order K^3 operations, relative to e-BH which only requires K operations. ♥ [David: I didn't have time to check exactly what the algorithm is, but if K is not super-large in our examples perhaps it's worth trying it out? Assuming that the algorithm is simple to implement or that code is available, of course.]

4 Results

Table 1 Rejected hypothesis by the eBH procedure with level 0.05

Outcome	Treatment (time on. . .)	e-value (E^{c1})
Low self-esteem (self-assessed)	internet	37,790
Depressive symptoms (self-assessed)	social media	1,851
High peer problems (parent-assessed)	internet	757
Depressive symptoms (self-assessed)	internet	396

We apply our multiverse analysis framework to the study of the effect of technology on teenager well-being in the 2015 survey of the Millennium Cohort Study (MCS). The MCS a socio-economic and health survey tracking a representative sample of children born in the UK around year 2000 throughout their life. The 2015 data that we use has a rather large sample size, $n = 11884$, and features several measures of well-being among teenagers and technology use.

An earlier analysis of this dataset in [3] concluded to no significant association between technology and teenager well-being. To assess significance of effects across outcomes and treatments, the authors used the Specification Curve Analysis [1] method which has serious shortcomings as briefly discussed in Section 1. When applied to MCS dataset, our framework leads to conclusions opposite to those in [3].

We consider $L = 8$ outcomes: self-assessed depressive symptoms (Mood and Feelings Questionnaire) and self-esteem (Rosenberg scale), as well as parent-assessed total difficulties, emotional problems, conduct problems, hyperactivity/inattention, peer problems and pro-sociality (all Strengths and Difficulties Questionnaire). There are $p_x = 5$ treatments: times spent watching TV, playing video games, using the internet and using social media, and owning a PC. In all regressions, we include $p_z = 14$ control covariates: sex, age, BMI, (self-reported) educational motivation, mother’s ethnicity, (self-reported) closeness to parents, presence of natural father, time spent with primary caretaker (PC), PC’s word activity score, PC’s employment status, own longstanding illness, PC’s psychological distress, number of siblings and household income.

The data were prepared as described in [2]. In particular, all treatment variables but TV usage were treated as continuous variables, normalized such that 0 means “no usage” and 1 means maximum reportable usage. TV usage was discretized into low, medium and high usage to boost the monotonic association with treatments. Outcomes variables are binary and were obtained by binning originally continuous data with validated cutoffs for abnormal behavior. See [2] (Section S4) for further details.

Following our proposed framework, we fit a logistic regression for each outcome and compute e-values for each outcome-treatment pair and apply the eBH procedure. In particular, we use e-variable E^{c1} that exhibits largest power in our simulations. We report in Table 1, the hypotheses rejected by the e-BH procedure with level 0.05. Alternative, less powerful, choices of e-variables in Section 3.1 lead to no rejection in this dataset. In total, four hypotheses are rejected by our framework suggesting a significant effect of technology on teenager well-being contrary to the original study in [3].

5 Discussion

Declarations

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Ethics approval. Not applicable.

Consent. Not applicable.

Data.

Code availability.

Author’s contribution statements. All authors contributed equally to this work.

Appendix A Section title of first appendix

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